PHYS 206: General Physics IV **KOÇ UNIVERSITY** **Spring 2026 Semester**

Semester: Spring 2026
Lecture Hours: MoWe 10:00-11:10
Room: SCI 103

Instructor: Alper Kiraz	**Coordinator:** Nazmi Yılmaz
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Off. Hr.: We 14:30-15:40 or by appointment	**Off. Hr.:** Mo TBA

Course Description: The nature and propagation of light, geometric optics and optical instruments, interference, diffraction, relativity, photons electrons and atoms, the wave nature of particles, quantum mechanics, atomic structure, molecules and condensed matter, nuclear physics, particle physics and cosmology.

Textbook: University Physics by H.D. Young and R.A. Freedman, 15th Edition, Addison-Wesley (2012). Available at the bookstore.

Week		**Subject**	**Week**		**Subject**
1	Feb. 9	Electromagnetic Waves (Ch.32) / The Nature and Propagation of Light (Ch. 33)	9	Apr. 6	Particles Behaving as Waves (Ch. 39)
2	Feb. 16	The Nature and Propagation of Light (Ch. 33)	10	Apr. 13	Quantum Mechanics (Ch. 40)
3	Feb. 23	Geometric Optics (Ch. 34)	11	Apr. 20	Quantum Mechanics (Ch. 40)
4	Mar. 2	Interference (Ch. 35)	12	Apr. 27	Atomic Structure (Ch. 41) / MT 2 (Ch.s 37-40)
5	Mar. 9	Diffraction (Ch. 36)	13	May 4	Molecules and Condensed Matter (Ch. 42)
6	Mar. 16	**Spring Break**	14	May 11	Nuclear Physics (Ch. 43)
7	Mar. 23	Relativity (Ch. 37) / MT 1 (Ch.s 32-36)	15	May 18	Particle Physics and Cosmology (Ch. 44)
8	Mar. 30	Photons: Light Waves Behaving as Particles (Ch. 38)		1-12 June	**Comprehensive Final Exam**

Grading: Midterm1: 22%, Midterm2: 22%, Final: 27%, Labs: 18% (3% per experiment), Homework: 6%, Attendance: 5%

Attendance Policy: Students are required to attend all lectures in person. Attendance will be recorded either with a physical sheet of paper circulated in class, on which you will be asked to write your names or through Learn Hub.

Make-up Policy: Students are expected to attend all lectures, laboratory experiments, and problem sessions (PS's). If a student misses a Midterm, the Final and a Laboratory Session and has a legitimate excuse approved by the University (s/he will be given an **all-inclusive make-up exam** on the official make-up date as scheduled by the Registrar's Office. Quizzes do not have makeups, instead, those you missed with a valid excuse will not be counted with the remaining ones weighing a greater amount.

Laboratory Attendance

Students are required to attend all scheduled laboratory experiments. Make-up laboratories are very reluctantly given only with a university approved medical excuse, and if given, will always be harder than the original laboratories. Students should not plan to take make-ups.

Before and during the laboratory experiment

Read the experiment manual before coming to the laboratory. Adhere to the laboratory time limit.
Do not forget to get your data checked and confirmed by the laboratory instructor before leaving the laboratory.
After completing the experiment, clean up your setup and properly turn off all the electronics. Make sure to show your assistant that there are no missing equipment pieces. You are not allowed to leave before the laboratory instructor's approval.
Taking any equipment out of the laboratory is an offense and may result in disciplinary action. Moreover, you will be financially responsible to replace all the missing equipment.

Laboratory Reports

Each Student should write a report of the experimental work and submit their report on LearnHub before the next experimental work.
The laboratory report should include the following items numbered in the specified format:
Introduction section including the objective of the experiment. (5 points)
Theory section including the theoretical background and the equations related to the experi-ment. (5 points)
Experimental setup section including a brief procedure of the experiment. (5 points)
Data analysis section including the data in tabular form and the plots with suitable titles, units, and scales on both coordinate axes. (40 points)
Discussion section including detailed answers to the questions. (40 points)
Conclusion section including conclusions drawn from the experiment. (5 points)

Tentative Laboratory Schedule

	Group1	Group2
Week 1- Feb.09	**No Lab**	
Week 2 Feb. 16	**Lab Orientation**	
Week 3 Feb. 23	Geometric Optics	
Week 4 Mar. 02	No Lab	
Week 5 Mar. 09	Young's Double Slit	
Week 6 Mar. 16	No Lab	
Week 7 Mar. 23	**Spring Break**	
Week 8 Mar. 30	Michelson Interferometer	No Lab
Week 9 Apr. 06	No Lab	Michelson Interferometer
Week 10 Apr. 13	Photoelectric Effect	No Lab
Week 11 Apr. 20	No Lab	Photoelectric Effect
Week 12 Apr. 27	Bohr Theory of Hydrogen-The Rydberg Constant	No Lab
Week 13 May 04	No Lab	Bohr Theory of Hydrogen-The Rydberg Constant
Week 14 May 11	Determination of e/m	No Lab
Week 15 May 18	No Lab	Determination of e/m